BUSINESS INTELLIGENCE SOLUTION

Multi-Specialty Medical Group

Project Documentation & Analysis Report

15	3	$6.13M
Doctors	Hospitals	Total Revenue

1. Project Overview

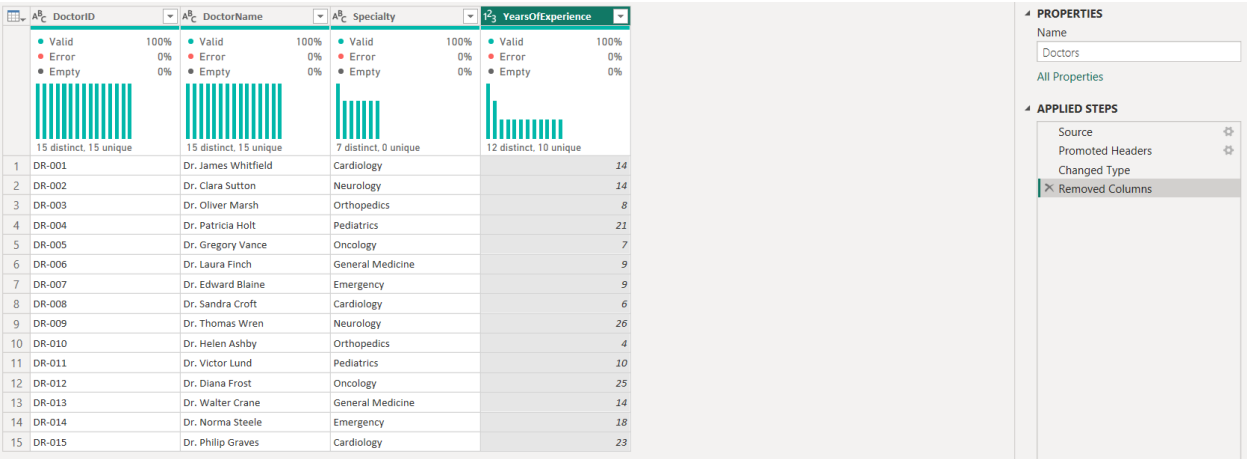
This project focused on developing a comprehensive, high-fidelity Business Intelligence solution for a multi-specialty medical group. The primary objective was to transform raw operational data into actionable insights to:

- Monitor financial health across hospitals and specialties
- Evaluate departmental and doctor-level performance
- Analyze patient satisfaction drivers and quality metrics
- Implement row-level security (RLS) for hospital-specific data access

2. Data Cleaning

2.1 Doctors Table

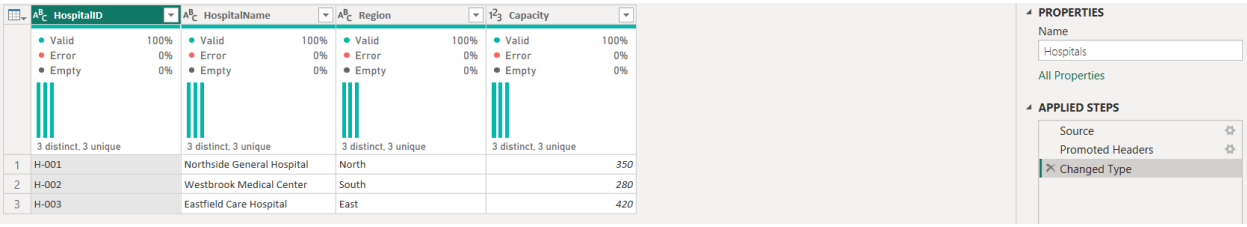
- Removed the Hospital ID column, connection is handled through the Visits fact table in the star schema model.



	DoctorID	DoctorName	Specialty	YearsOfExperience
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	15 distinct, 15 unique	15 distinct, 15 unique	7 distinct, 0 unique	12 distinct, 10 unique
1	DR-001	Dr. James Whitfield	Cardiology	14
2	DR-002	Dr. Clara Sutton	Neurology	14
3	DR-003	Dr. Oliver Marsh	Orthopedics	8
4	DR-004	Dr. Patricia Holt	Pediatrics	21
5	DR-005	Dr. Gregory Vance	Oncology	7
6	DR-006	Dr. Laura Finch	General Medicine	9
7	DR-007	Dr. Edward Blaine	Emergency	9
8	DR-008	Dr. Sandra Croft	Cardiology	6
9	DR-009	Dr. Thomas Wren	Neurology	26
10	DR-010	Dr. Helen Ashby	Orthopedics	4
11	DR-011	Dr. Victor Lund	Pediatrics	10
12	DR-012	Dr. Diana Frost	Oncology	25
13	DR-013	Dr. Walter Crane	General Medicine	14
14	DR-014	Dr. Norma Steele	Emergency	18
15	DR-015	Dr. Philip Graves	Cardiology	23

2.2 Hospital Table

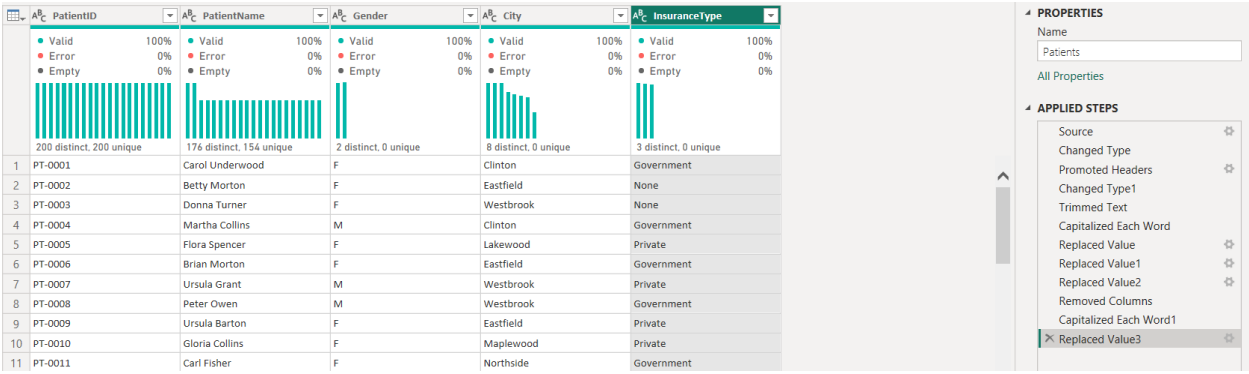
- No cleaning required. Data was already complete and well-structured.



	HospitalID	HospitalName	Region	Capacity
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	3 distinct, 3 unique	3 distinct, 3 unique	3 distinct, 3 unique	3 distinct, 3 unique
1	H-001	Northside General Hospital	North	350
2	H-002	Westbrook Medical Center	South	280
3	H-003	Eastfield Care Hospital	East	420

2.3 Patients Table

1. Promoted first row as column headers.
2. Trimmed leading whitespace from the PatientName column.
3. Capitalized each word in PatientName; standardized Gender values to 'Female' and 'Male'.
4. Replaced empty City values with 'Unknown'.
5. Removed the DateOfBirth column due to erroneous date entries and irrelevance to analysis goals.
6. Standardized InsuranceType values (capitalized) and replaced null entries with 'None'.



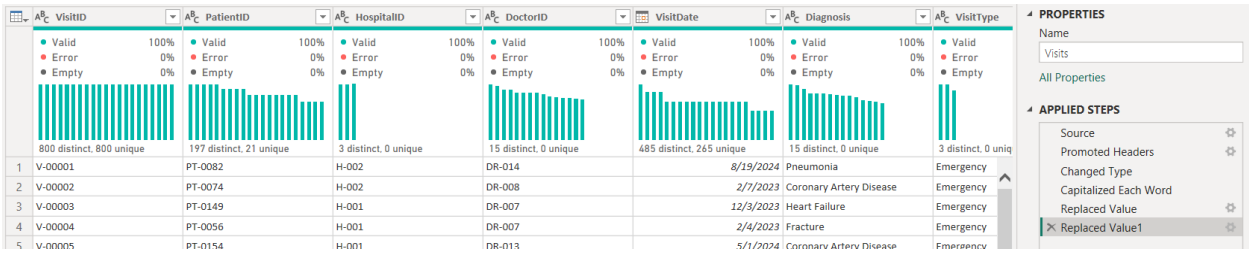
	PatientID	PatientName	Gender	City	InsuranceType
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	200 distinct, 200 unique	176 distinct, 154 unique	2 distinct, 0 unique	8 distinct, 0 unique	3 distinct, 0 unique
1	PT-0001	Carol Underwood	F	Clinton	Government
2	PT-0002	Betty Morton	F	Eastfield	None
3	PT-0003	Donna Turner	F	Westbrook	None
4	PT-0004	Martha Collins	M	Clinton	Government
5	PT-0005	Flora Spencer	F	Lakewood	Private
6	PT-0006	Brian Morton	F	Eastfield	Government
7	PT-0007	Ursula Grant	M	Westbrook	Private
8	PT-0008	Peter Owen	M	Westbrook	Government
9	PT-0009	Ursula Barton	F	Eastfield	Private
10	PT-0010	Gloria Collins	F	Maplewood	Private
11	PT-0011	Carl Fisher	F	Northside	Government

PROPERTIES
Name
Patients
All Properties

APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Trimmed Text
Capitalized Each Word
Replaced Value
Replaced Value1
Replaced Value2
Removed Columns
Capitalized Each Word1
Replaced Value3

2.4 Visits Table

7. Capitalized each word in the VisitType column.
8. Replaced null values in LengthOfStay with 0 to ensure analytical compatibility.
9. Replaced the invalid score of 99 in SatisfactionScore with 5 to align with the valid 0–5 range.



	VisitID	PatientID	HospitalID	DoctorID	VisitDate	Diagnosis	VisitType
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
	800 distinct, 800 unique	197 distinct, 21 unique	3 distinct, 0 unique	15 distinct, 0 unique	485 distinct, 265 unique	15 distinct, 0 unique	3 distinct, 0 uniq
1	V-00001	PT-0082	H-002	DR-014	8/19/2024	Pneumonia	Emergency
2	V-00002	PT-0074	H-002	DR-008	2/7/2023	Coronary Artery Disease	Emergency
3	V-00003	PT-0149	H-001	DR-007	12/3/2023	Heart Failure	Emergency
4	V-00004	PT-0056	H-001	DR-007	2/4/2023	Fracture	Emergency
5	V-00005	PT-0154	H-001	DR-013	5/1/2024	Coronary Artery Disease	Emergency

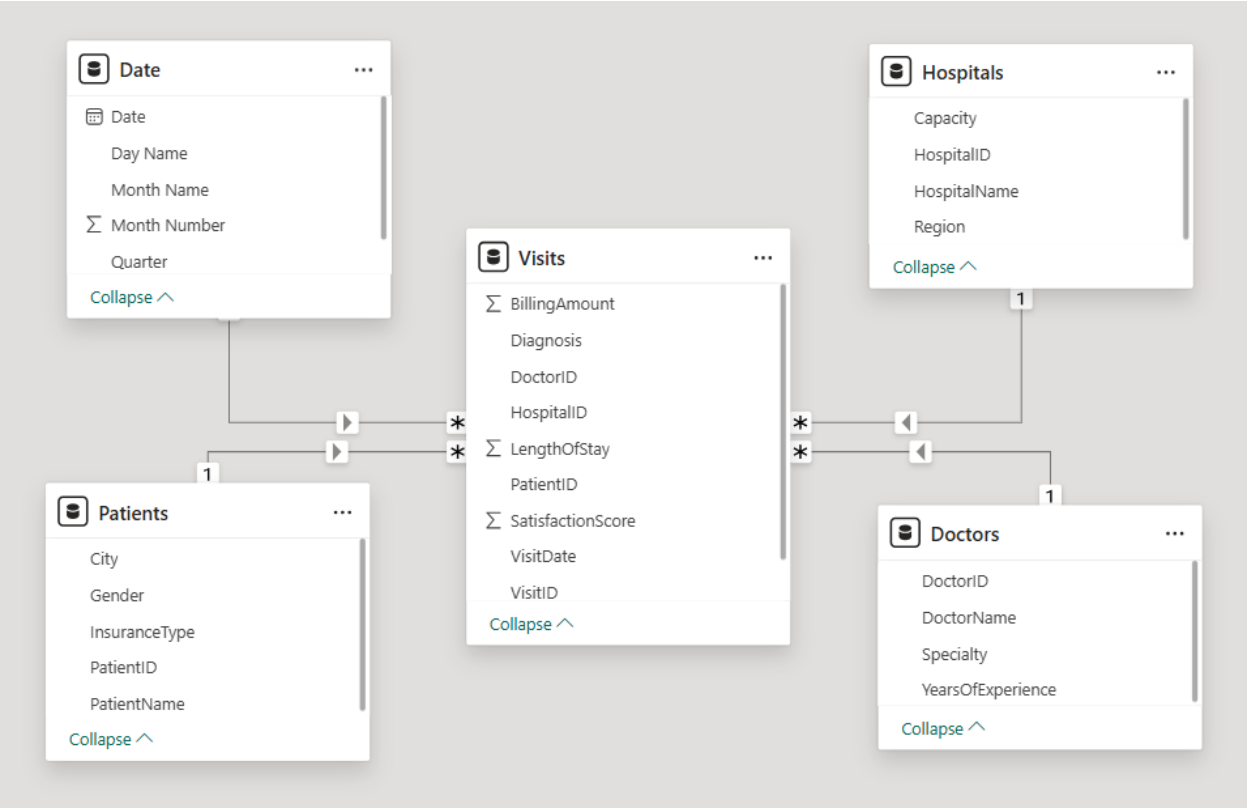
PROPERTIES
Name
Visits
All Properties

APPLIED STEPS
Source
Promoted Headers
Changed Type
Capitalized Each Word
Replaced Value
Replaced Value1

3. Data Modeling

A star schema was implemented with Visits as the central fact table, connected to four dimension tables:

Table	Type	Key Columns
Visits	Fact	VisitID, PatientID, DoctorID, HospitalID, VisitDate
Doctors	Dimension	DoctorID, DoctorName, Specialty, YearsOfExperience
Patients	Dimension	PatientID, PatientName, Gender, City, InsuranceType
Hospitals	Dimension	HospitalID, HospitalName, Region, Capacity
Date	Dimension (DAX)	Date, Year, Month, Quarter, Day Name



3.1 DAX Date Table

The Date dimension was generated using the following DAX expression:

```
Date = ADDCOLUMNS(
    CALENDARAUTO(),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", FORMAT([Date], "Q"),
    "Day Name", FORMAT([Date], "dddd") )
```

4. Row-Level Security (RLS)

RLS was configured to restrict each hospital manager's view to data from their assigned hospital only.

4.1 Security Table

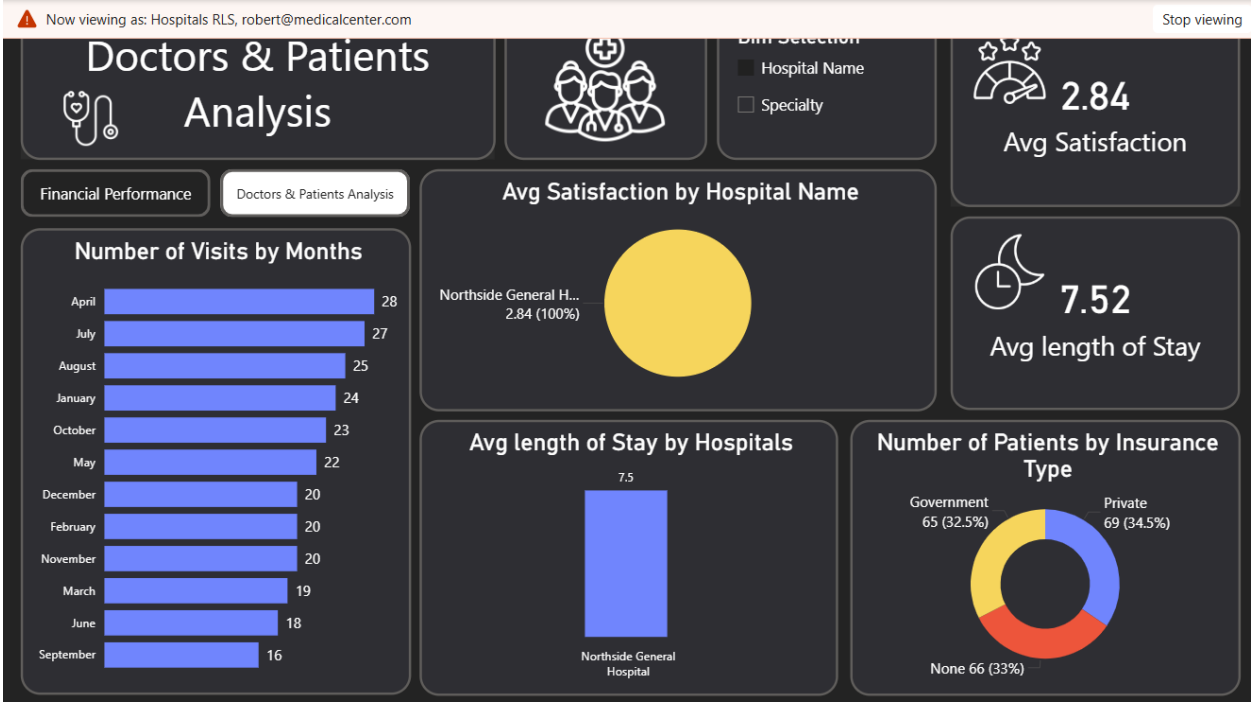
User Email	Assigned Hospital
robert@medicalcenter.com	Northside General Hospital
sandra@medicalcenter.com	Westbrook Medical Center
patricia@medicalcenter.com	Eastfield Care Hospital

4.2 RLS DAX Rule

The following filter was applied on the Security Table to enforce row-level access:

```
[User Email] = USERPRINCIPALNAME()
```

4.3 RLS Test



5. Questions & Answers

5.1 Revenue & Financial Performance

Question	Answer
Total revenue across all hospitals?	$6.13 M
Which hospital generates the highest revenue?	Eastfield Care Hospital, followed by Northside General and Westbrook Medical
Month-over-month trend — any drops or spikes?	Current month: $275K vs. previous: $321K (drop). Notable dip also in January 2024.
Revenue breakdown by visit type?	Outpatient: 35.93% | Emergency: 32.8% | Inpatient: 31.27%
Which doctor specialty drives the most revenue?	Cardiology
Top-performing doctor by total revenue billed?	Dr. Philip Graves

5.2 Patient & Quality Metrics

Question	Answer
Overall average patient satisfaction score?	2.95 / 5
Proportion of insured patients?	Private: 34.5% | Government: 32.5%
Average length of stay for inpatient visits?	7.35 days — minimal differences between hospitals
Any months with significantly higher patient visits?	August (highest: 79 visits) vs. February (lowest: 52 visits)

5.3 Doctor & Operational Performance

Question	Answer
Which specialty has the highest avg patient satisfaction?	Orthopedics
Relationship between experience and satisfaction scores?	Yes — doctors with 8 years of experience show the highest average satisfaction rate

6. Recommendations

1

Optimize Revenue Stream Allocation

Outpatient services ($2.20M, 35.93%) and Emergency visits ($2.01M, 32.8%) are the primary revenue drivers. Expand outpatient clinic hours or emergency service capacity to capitalize on these high-performing visit types.

2

Standardize Patient Satisfaction Across Hospitals

Eastfield Care Hospital leads with a score of 3.17, while Northside General (2.84) and Westbrook Medical (2.85) lag behind. Conduct a qualitative review at Eastfield to identify best practices and replicate them across the other facilities.

3

Review Resource Allocation for Low-Revenue Specialties

Oncology and Orthopedics currently show the lowest total revenue. Analyze whether these departments are under-resourced or face low regional demand. If demand is naturally low, consider consolidating into a single Centre of Excellence.

4

Leverage Top-Performing Talent

Dr. Philip Graves is identified as the top-performing doctor by revenue. Implement a peer-mentorship program where top performers share patient care techniques, as the scatter analysis suggests soft skills may drive satisfaction more than years of experience.

5

Address Recent Revenue Volatility

MoM revenue shows a 14.31% decline (from $321K to $275K). Investigate the October 2024 dip in the revenue trend, check for seasonal staffing shortages or billing delays during that period to prevent a recurring year-end slump.